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28 June 1976
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MILITARY SPECIFICATION

ELECTRIC GENERATING SYSTEM, VARIABLE SPEED CONSTANT FREQUENCY, AIRCRAFT, GENERAL SPECIFICATION FOR

This specification has been approved by Naval Air Systems Command, Department of the Navy.

1. Scope. - This specification establishes the requirements for a 3 phase, 4 wire, 400 Hz Variable Speed Constant Frequency (VSCF) electric power system. Such a system is defined as one in which the mechanical input power of a variable speed shaft(s) is converted to electrical output power at a constant frequency by means of variable speed generator(s) and a static converter(s). A VSCF channel is defined as the equipment required to convert, regulate and monitor the power from one variable speed shaft to the point of regulation. A system shall consist of all the channels installed on a particular airplane and may comprise either a single channel or two or more channels operating in the isolated or parallel configuration,

2. APPLICABLE DOCUMENTS

2.1 General. - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS AND STANDARDS

MIL-W-5088	Wiring, Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Aircraft General Specification for
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance
MIL-C-6021	Castings, Classification and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document by letter.

FSC 6115

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MIL-S-7742	Screw Threads, Standards Aeronautical
MIL-G-21480	Generator System, Single Generator, Constant Frequency, Alternating Current, Aircraft,; General Specification
MIL-T-22520	Tool, Crimp Type for Contacts of Electric Connectors
MIL-L-23699	Lubricating Oil, Aircraft, Turbine, Engine Synthetic Base
MIL-G-27617	Grease, Aircraft, Fuel and Oil Resistant
MIL-R-39015	Resistors, Variable, Wire-Wound, (Lead screw actuated Established Reliability
MIL-E-81910	Electric Power and Control Equipment, Aircraft, Testing; General Specification For
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-481	Configuration Control-Engineering Changes, Deviations and Waivers
MIL-STD-595	Aeronautical Standard Notice For Federal Std. No. 595 Colors
MIL-STD-704	Electric Power, Aircraft, Characteristics And Utilization of
MIL-STD-781	Reliability Tests: Exponential Districation
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing Of

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MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS3376	Nut, Self-Locking, Reduced Height, Steel, 450 °F, 12 Points, Captive Washer
MS 90415	Nut, Self-Locking, Steel, 160KSI, 450°F, 12 Points, Captive Washer

NAVAL AIR SYSTEM COMMAND REQUIREMENTS

AR-5	Microelectronic Devices Used In Avionics Equipment, Procedures For Selection And Approval
AR-8	Versatile Avionic Ship Test System, Avionic System Compatibility, General Requirements For
AR-9	VAST Test Programs, General Requirements For
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements For

STANDARDS

ASA	
C42.05	General Terms

2.2 Availability of Documents. - When requesting Specifications, and drawings, refer to both title and number. All requests should be made to the procuring activity or as directed by the contracting officer.

2.3 Precedence. - The contract shall have precedence over any specification. Whenever the requirements of this specification and the detail specification or standard conflict, the requirements of the detail specification or standard shall govern. Any deviation from this specification or from subsidiary specifications or standards, where applicable, are subject to approval in writing by the Qualifying Activity.

3. REQUIREMENTS

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3.1 Qualification. - The Naval Air Test Center (SY60), Patuxent River, Maryland 20670 is the qualifying activity for equipment submitted under this specification. Qualification under this specification is required. The unit furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein and approved for listing on the applicable Qualified Products List unless agreement is obtained in writing from the qualifying activity that no qualified product is practical for the application. In the latter case, a product which has been tested and has qualified to a detail specification accepted in writing by the Qualifying Activity may be utilized. In either case, each unit delivered shall fully meet all the requirements of the applicable detail specification and shall be capable of successfully completing the tests required by the detail specification for acceptance and for any one of the qualification samples.

3.2 System Capacity (Rating). - Each channel within a system shall provide two sources of power; one (the main power source), for general distribution to aircraft electric equipment; and two, a power source (integral control power) for use within the channel for regulating, controlling, and monitoring the power generated and, in addition, for use outside the channel for powering feeder and bus contractors.

3.2.1 Rating of the Main Power Source. - The channel rating shall be as stated by the detail specification and shall be based on 115/200 volt (rms), 400 Hz, 3 phase, 4 wire electrical power. All ratings are at the point of regulation which is the point at which each channel is connected to its 400 Hz load bus. The length, configuration, and impedance of all feeders with which these ratings must be met shall be as defined in the detailed specification. All ratings defined in the following subparagraphs are applicable over the generator rpm range and environmental conditions defined herein. If the channel is defined with a slash rating it shall be capable of meeting all requirements of this specification at either rating except overload and short circuit capacity. The overload and short circuit capacity are based on the lowest specified continuous duty rating.

3.2.1.1 Continuous Capacity. - The channel shall be capable of continuously delivering the rated capacity defined by the detailed specification for load power factors of 0.75 lagging to 0.95 lagging.

3.2.1.2 Overload Capacity. - The channel shall be capable of delivering current into a 150% rated load, at any power factor from 0.75 lagging to 0.95 lagging for 5 minutes. It shall also be capable of supplying current into a 200% rated load, at any power factor from 0.75 lagging to 0.95 lagging, for 5 seconds.

3.2.1.3 Motor Starting and Running Capacity. - With an initial channel load from zero percent or rated load at any power factor from 0.75 lagging to 0.95 lagging, the channel shall be capable of supplying current for 5

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seconds to a simulated locked rotor motor representing 80 percent of rated load at 0.4 lagging power factor. The requirements of figure 1 shall be met.

3.2.1.4 Short Circuit Capacity. - The channel shall produce a minimum of 300 percent of rated phase load current on at least one phase into single phase, two phase, and three phase line-to-line and line-to-neutral short circuit for 5 seconds. When operated in parallel, the short circuit current capacity of the system shall be 250 to 400 percent of the total rated phase load capacity of the system on at least one phase for the same fault conditions and time periods required of a channel.

3.2.2 Integral Control Power. - The channel shall provide control power independent of the primary AC output from the converter and from all sources external to the channel. The integral control power capacity shall be adequate for channel control and protection and, in addition, shall be capable of furnishing DC power with the following characteristics to contactors external to the system at generator rpm's of from 90 percent of the minimum rated rpm to 110 percent of maximum rated rpm: 20 volts minimum voltage, 30 volts maximum voltage, 2 volts maximum ripple amplitude with a continuous output capacity of up to 5 amps, a 5 second capacity of 7.5 amps minimum, and a short circuit capacity of 10 amps minimum. This power shall be available at the end of feeders through which there is a one volt drop when 5 amps is being delivered.

3.3 System Performance. - All characteristics specified herein are at the point of regulation and are applicable over the generator rotor rpm range (or any combination of generator rotor rpm's if a multi-channel system) and environmental conditions specified herein. No tripping of the system shall be caused by build up, load switching, generator input rpm acceleration or deceleration, or external fault application, or the clearing of a fault external to the system the power requirement for which does not exceed system capacity defined by paragraph 3.2.

3.3.1 Output Voltage. - In each instance below requiring a protective device, reset shall not be automatic. Reset shall be by means of the manual control switch (see paragraph 3.4.1.3) only.

3.3.1.1 Voltage Regulation. - The transient and steady state line-to-neutral voltages of the main power output shall remain within the limits of Figure 1. These requirements shall be met for all balanced loads from no load to loads as defined by 3.2 and for unbalanced loads within the rating of the channel, where the differences in the per phase load currents do not exceed 1/3 of the phase current rating. In addition, the line-to-line voltages, under steady state conditions shall remain within the limits of 196 to 202 volts. Any condition imposed on the channel resulting in less than 105 volts, line-to-neutral, at the point of regulation on any phase shall be cleared or the line conductor opened and the generator deenergized within a period of 3.5 to 5 seconds. This protective function shall not occur when the line-to-neutral voltage exceeds 109 volts. Protective functions within the system shall open the line contactor and deenergize

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the channel prior to any individual phase voltage exceeding the limits defined by Figure 2. The integral control voltage shall not exceed the limits specified by paragraph 3.2.2.

3.3.1.2 Wave Form. - From no load to full load, the phase voltage wave shall have a crest factor of 1.41 ± 0.07, the distortion factor for the phase voltage waveform shall not exceed 0.05, nor shall the limits of the ac distortion spectrum exceed the envelope shown in Figure 3. For loads from full load to 150 percent, the distortion factor shall not exceed 0.10. The dc content of the output voltage shall not exceed ± 0.10 volts under any operating condition up to full load. Protective functions within the system shall operate to open the line contactor and deenergize the channel when the components of frequencies greater than 400 Hz exceed Figire 4 for 3.5 to 5 seconds or when the dc content on any phase exceeds 1.0 volt second above an ultimate trip point of 0.5 volts dc.

3.3.1.3 Voltage Phase Difference. - The voltage phase difference shall be within $120^{o} \pm 2^{o}$.

3.3.2 Output Frequency. -

3.3.2.1 Frequency Regulation. - One of the following classes of the regulation limits is to be indicated by the detailed specification. The regulation limits indicated shall pertain under all loading conditions (transient and steady state) specified.

Class A. Frequency limits shall remain within 25 parts per million of 400 Hz (400.00 ± 0.01 Hz).

Class B. Frequency limits shall remain within 250 parts per million of 400 Hz (400.00 ± 0.10 Hz).

Class C. Frequency limits shall remain within 2500 parts per million of 400 Hz (400.00 ± 1.00 Hz).

A protective function within the channel shall operate to open the line contactor and deenergize the generator within a period of 0.1 to 0.5 seconds after the output frequency has varied outside the limits of 400 ± 2 Hz. There shall be no automatic reset. Reset shall be by means of the manual control switch (see paragraph 3.4.1.3) only.

3.3.2.2 Frequency Modulation. Modulated frequency deviation shall not exceed the limits specified in Figure 5.

3.3.3 Efficiency. - The minimum channel efficiency and maximum heat rejection for each channel component shall be as specified by the detail specification at rated load and power factor range. Losses at any specified generator rpm shall not be greater at loads less than rated than they are at rated load.

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3.3.4 Parallel Operation. - It shall be possible to parallel two or more channels of the same type and manufacture on a continuous basis if required by the detail specification.

3.3.4.1 Load Division. - The KVA output of each line-to-neutral phase of a channel operated in parallel shall not deviate from the average KVA output of all parallel channels by more than 15 percent of the rating of that channel at any instant or by more than 7.5 percent of the rating of that channel for any one minute period. These limits apply to all steady state conditions within the systems rated capacity. When channels of different ratings are paralleled, the load sharing shall be proportional to their respective ratings within the above limits of this paragraph.

3.3.4.2 Short Circuit Capacity. - The short circuit current capacity shall be in accordance with paragraph 3.2.1.4.

3.3.4.3 Protection. - When two or more channels are operating in parallel, protection shall be provided as follows:

(a) When parallel system output parameters exceed the protective limits defined for single system operation, protective functions shall operate to restore normal characteristics of electrical power to the loads.

(b) To prevent tripping all channels either simultaneously or in cascade as the result of any single malfunction.

(c) To prevent damage to remaining satisfactorily operating channel(s) as the result of any single channel malfunction.

(d) To selectively isolate the malfunctioning channel or short circuited bus with a minimum reduction of system capacity and minimum interruption of bus power.

3.3.5 Split-Synchronized Operation. - When dictated by the detail specification, the channel shall operate isolated but synchronized (split-synchronized) with another channel or a constant speed electric generating system having a frequency between 390 and 410 Hz. While synchronized, the output waves per phase shall be within 15 degrees of each other under all normal operating conditions including load application and removals. Protection shall be the same as required for a parallel system, paragraph 3.3.4.3.

3.3.6 No-Break Power Transfer. - When dictated by the detail specification, the channel shall be capable of automatically transferring loads from, and to, another electric system having a frequency between 390 and 410 Hz, and voltage conforming to MIL-STD-704 without any interruption of power to electric loads and without creating voltage transients exceeding those caused by the application of removal of full load.

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3.3.7 Maximum Fault Torque. - The channel mechanical demand from its driving sources shall not exceed 400% of that required at rated load for any combination of 400 Hz short circuit or fault impedance.

3.3.8 Electromagnetic Interference. - The system, as well as each channel, shall meet the requirements of MIL-E-81910. Conducted radio noise requirements pertain to the 400 Hz feeders only. Radiated noise requirements pertain to all channel components, including interconnecting wiring.

3.4 Design and Construction Requirements. - All channel components shall conform to the applicable detail specification.

3.4.1 General Design and Construction Requirement. - All system components shall conform to the applicable detail specification.

3.4.1.1 Acceptable Design. - Design of each unit shall be consistent with good aircraft accessory practices. Careful consideration shall be given in the design of the unit to simplify servicing, inspection (testing), and overhaul. The necessity of using special tools and fixtures for maintenance and overhaul shall be kept to a minimum.

3.4.1.2 Environmental Requirements. - All components shall operate in accordance with this specification when subjected to the tests of MIL-E-81910 under the temperature-altitude conditions and with the cooling means designated by the detailed specification. (See paragraph 5.3 of MIL-E-81910 for required inclusions in detail specifications).

3.4.1.3 Manual System Control. - Provisions shall be made for manual control of the system, to be accomplished by means of one switch per channel. This switch shall be a standard switch and shall perform the following control functions in the positions indicated.

NORMAL - Each system automatically energizes its bus when its electrical characteristics are within prescribed limits. Each system shall automatically parallel or synchronize with other system(s), as required by the detail specification. There shall be no interruption of power to load equipment during paralleling or synchronizing. Warning light "ON" when power bus is not energized. Protective circuits operative.

OFF-RESET - The system is electrically de-energized and main line contactor open. Protective elements reset to normal (See paragraph 3.4.3.1). Warning light "ON".

TEST - (Momentary) Channel energized with main line contactor open. Protective circuits, operative. Warning light "OFF" if power is within protective limits (paragraph 3.4.3.1), otherwise, light is "ON".

3.4.1.4 Interchangeability. - Each article of the same part number manufactured in accordance with this specification shall be physically and functionally interchangeable with like articles bearing the same contractor part number.

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3.4.1.5 Workmanship.- Workmanship shall be in accordance with high grade manufacturing practices for aircraft accessories and equipment. The units shall be free from dirt, sand, metal chips, machining compounds, and other foreign matter. All machined surfaces shall have a smooth finish and all details of manufacture, including the preparation of parts and accessories, shall be in accordance with good practice for high quality electric equipment. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts, plating, coatings, riviting, clearance between soldered connections, and ruggedness.

3.4.1.6 Shock Mounts. - Shock mounts (or vibration isolators) shall not be used.

3.4.1.7 Electron Tubes and Semi-Conductors. - Transistors and diodes shall be chosen, applied, and reported as outlined in Specification MIL-STD-454. The component report must be submitted to the qualifying activity for review and approval prior to qualification testing. Electron tubes of either the vacuum or gas filled types shall not be used.

3.4.1.8 Electric Connections. -

3.4.1.8.1 External Electric Connections. - The equipment shall be so designed that external connections and wiring can be made in accordance with MIL-W-5088. Equipment structure shall not be used as a current path except for radio noise shielding.

(a) Connectors. Connectors shall be polarized or sized so as to prevent their interchange during equipment installation.

(b) Terminal Block Design. Where terminals are used for the external connection of components they shall be of the stud-type and shall be designed so that the current is conducted by means of surface to surface contact and not through the stud threads. All studs shall be steel, corrosion resistant, C-34 Rockwell minimum hardness, and shall be of sufficient length to accommodate two MIL-T-7928 lugs. All nuts by which external wiring is attached to engine or nacelle mounted channel components shall conform to MS90415. Nuts used on terminals of channel components mounted elsewhere in the aircraft shall conform to either MIL-STD-889 or MS90415. There shall be no dielectric material in the compression buildup. The terminal block shall be so designed that it can be removed and replaced on the component without the necessity for rebrazing or soldering. Barriers affording a positive separation of leads and terminals shall be provided on the terminal block. Durable, reusable, non-conductive terminal

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covers shall be provided. Terminal designations shall be durably, legibly, and prominently marked on the terminal block.

3.4.1.8.2 Internal Electric Connections. - Internal connection shall be made in accordance with MIL-E-5400 and MIL-STD-454.

3.4.1.9 Threaded Parts. - All threads shall conform to MIL-S-7742. All internal or external parts which are threaded shall be positively locked. Accidental loosening of threaded elements shall be prevented by self locking nuts, safety wiring, staking, or other approved methods. Staking shall not be used. All elements externally screwed into the unit (screws, bolts, caps, plugs, filter, fittings, etc.) are to be screwed into a solid steel member and are to be secured from "backing out" by safety wire. Coiled members, inserts are not to be used. Cadmium plated screws are not to be used. All externally accessible bolts and screws with the exception of those securing electric receptacles, are to be 1/4 inch minimum in diameter for engine mounted equipment.

3.4.1.10 Adjustments. - No external adjustment means shall be provided and no adjustment shall be required during installation of channel or any channel component. All adjustments are to be made at the time of manufacture or overhaul. If devices for making adjustments are installed within components, suitable locking and sealing means shall be provided to indicate when an unauthorized adjustment has been made.

3.4.1.11 Potentiometers and Adjustable Resistors. - Sealed potentiometers conforming to specification MIL-R-39015 may be used in the design of these systems. The types of adjustable resistors shall not be used without specific written approval of the qualifying activity.

3.4.1.12 Operating Service Time. - Rotating equipment shall operate for a minimum of 2000 hours to first overhaul without maintenance other than inspection and addition of lubricant. Static system components shall operate for a minimum of 3000 hours to first overhaul without maintenance.

3.4.1.13 Weight. - The weight of channel components and the overhung moment of engine mounted channel components shall not exceed the limits defined by the detailed specification.

3.4.1.14 Reliability. - The channel shall have a minimum acceptable MTBF (Mean Time Between Failure) of 800 hours.

3.4.1.15 External Openings in System Components. - All air vents or case openings shall be protected by suitable means against passage of round objects of 3/16 inch or greater in diameter. Air openings are excepted if to be fitted to a blast tube.

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3.4.1.16 Color. - Equipment shall be furnished in a color conforming to FED-STD-595 color number 17875 except that as an alternate engine mounted channel components may be finished resulting in an aluminum or stainless steel color. The finishes shall be unaffected by environmental testing required by the detail specification.

3.4.1.17 Provision for Maintainability. - Built in test equipment, construction and packaging, and provisions for test points shall be as specified in Specification AR-10.

3.4.1.17.1 Compatibility with VAST. - Equipment shall be compatible with VAST (Versatile Automatic Shop Test System) and shall meet the requirements of Specification AR-9. If changes to the equipment are made which will affect the fault diagnosis procedure, changes to the existing test program shall be prepared as part of the equipment changes in accordance with Specification AR-9.

3.4.1.17.2 Overhaul. - Complete overhaul of each component in accordance with overhaul instructions approved by the qualifying activity shall not exceed the time specified by the detail specification when performed by personnel with the recommended skill level.

3.4.1.17.3 Maintainability Demonstration. - A maintainability demonstration consisting of a replacement action as defined by the detail specification for each component shall not exceed the time specified by the detail specification when performed by personnel with the recommended skill and without special tools.

3.4.1.18 Coolant. - Coolants, other than air, used in channel components shall conform to MIL-L-23699. Combined leakage of such coolants from all channel components shall not exceed 1 cc per hour when the unit is either operating or inoperative.

3.4.1.19 Lubrication and Bearings.

3.4.1.19.1 Grease Lubricated Bearings. - When grease lubricated bearings are used in component they shall be packed with Krytox-240AC* grease meeting Specification MIL-G-27617. The grease shall fill not less than 45% or more than 55% of the bearing voids and be equally distributed across the race width. Bearing rings, balls or rollers, and shields shall be made of AISI-440C stainless steel. The rings and rolling elements shall be heat treated to a Rockwell "C" scale hardness of 58 to 62. Bearing tolerances shall meet the AFBMA class ABEC-7. If bearing supports are non-ferrous, the bearing shall be inserted into a steel sleeve mechanically secured to the bearing supports. If bearing supports are made of steel, sleeves are not required. An axial retention flange confining the outer bearing ring and a positive method to prevent rotation of the outer bearing ring shall be provided. The bearing inner ring shall be provided with sufficient interference fit with the shaft to prevent rotation under operating speeds and temperatures.

* E. Dupont de Nemours & Company, Incorporated Trade Mark

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3.4.1.19.2 Oil Lubricated Bearings. - When oil lubricated bearings are used in any channel component, oil conforming to MIL-L-23699 is required.

3.4.2 Generator (See paragraph 6.2.13) Design and Construction. - In addition to the requirements of paragraph 3.4.1, the following pertain to the generator package:

3.4.2.1 Brushes. - No brushes shall be used.

3.4.2.2 Generator R.P.M.. - The RPM range over which the generator package shall operate is to be defined by the detail specification. The generator shall be capable of operation at 110 percent of the maximum rated input rpm for 5 minutes after having stabilized thermally while operating at full rated load and minimum input speed. Subsequent channel performance in meeting specification requirements shall not be impaired.

3.4.2.3 Corona. - The generator design shall be such that no problems occur due to corona under any operating condition specified herein.

3.4.3.4 Shear Section. - A shear section shall be provided so that the spline shaft mating with the engine pad will shear at torsinal load limits as shown in the detailed specification.

3.4.3.5 Input Shaft Disconnect. - A disconnect shall be incorporated which when activated shall allow the engine pad output shaft to rotate free of the generator. For oil cooled generator, activation shall be automatic and shall disconnect the generator prior to loss of case integrity or case temperature reaching the coolant flash point. For air cooled generators, the means of sensing an impending generator failure and the means of activating the decoupler shall be as designated by the detail specification.

3.4.2.6 Effects of Fluids. - The generator package shall show no deterioration due to wetting from fluids conforming to Specification JP-4 fuel, MIL-H-5606, MIL-L-23699, Skydrol 500, and aircraft washing compounds at normal operating temperatures.

3.4.2.7 Operating Position. - The normal mounting position for the generator shall be as shown by the detail specification. The unit shall operate continuously in the normal position $\pm 20^{o}$ rotation and up to five minutes out of every hour of operation in any other position.

3.4.3 Design and Construction of the Converter Package and Associated Control Components - All channel elements for the conversion, regulation, and monitoring of the output power shall be contained in the converter with the exception of the manual system control (paragraph 3.4.1.3) and sensing elements defined by the detail specification. The requirements

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of paragraph 3.4.1 pertain to the converter package in addition to the following. In addition, if the converter is installed as part of the generator, paragraph 3.4.2.6 also pertains (See paragraph 6.2.13).

3.4.3.1 Regulation and Protective Characteristics. - In addition to the regulating and protective functions required by paragraphs 3.3.1 and 3.3.2 the following functions are required. After the operation of a protective element, with the exception of the generator underspeed monitoring element as described below, the system shall not automatically reset. Reset of protective elements shall be by means of the manual control (paragraph 3.4.1.3).

3.4.3.1.1 Feeder Fault. - When the rms current in a fault in any feeder exceeds 25 amperes, a protective function within the channel shall immediately operate to open the line contactor and deenergize the generator. No fault outside the feeder zones shall cause the feeder fault protective element to function.

3.4.3.1.2 Generator Underspeed. - When the generator rpm drops below the minimum indicated by the detail specification, a monitoring function within the system shall operate to open the line contactor and deenergize the channel. No manual reset shall be required to resume normal operation of the channel when the generator rpm returns to within the specified limits.

3.4.3.1.3 Anti-Cycling. - A means to prevent cycling the channel when a fault exists shall be provided. The control assembly shall reset once after a fault upon activation of the control switch to the "RESET" position and if the fault remains the assembly shall trip when the channel control is placed in the "NORMAL" or "TEST" position and remain tripped until the control switch is reactivated to the "RESET" position.

3.4.3.2 Modular Construction. - Modular construction shall be employed in the construction of subassemblies made up of electronic components. For aircraft under Navy cognizance equipment shall be modularized in accordance with Specification AR-10 and microelectric modular assemblies shall meet the requirements of Specification AR-5.

3.4.3.3 Operating Position. - All system components, excluding the generator (see paragraph 3.4.3.7), shall meet the requirements of this specification while mounted in any position.

3.4.4 Installation Instructions. - The contractor shall pack with each channel component one printed copy of simple instructions, with illustrations and required diagrams covering its installation. These instructions shall be protected to withstand and remain completely legible under the humidity and fungus requirements specified herein for

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self cooled components. Prior to printing, two copies shall be furnished the qualifying activity for approval of content, presentation, and means of attachment.

3.5 Materials. - Materials used in the manufacture of channel components shall be of high quality, suitable for the purpose, and shall conform to applicable government specification. Materials conforming to contractor's specification may be used provided the specifications are approved by the qualifying government activity and contain provisions for adequate tests. The use of the contractor's specifications does not constitute waiver of government inspection.

3.5.1 Metals. -

3.5.1.1 Corrosion Resistance. - Materials shall be of a corrosion resisting type or suitably processed to resist corrosion.

3.5.1.2 Use of Magnesium. - Magnesium shall not be used without specific written approval by the qualifying agency.

3.5.1.3 Dissimilar Metals. - The use of dissimilar metals in contact with each other (particularly brass, copper, or steel in contact with aluminum, magnesium, or their alloys) shall be avoided. Where contact between dissimilar metals is unavoidable, they shall be suitably protected against electrolytic corrosion. When protection is used, it shall be a type that offers a low-impedance path to radio frequency current. Dissimilar metals are defined in Standard MS 33586.

3.5.1.4 Castings. - Castings shall be classified and inspected to design requirements in accordance with MIL-C-6021. All castings containing reinforced threaded holes shall be classified Class 1B, Grade A in areas within three radii of the hole centerline for the depth of the hole.

3.5.2 Non-Metallic Materials. - Non-metallic material used shall be flame resistant and shall not support combustion.

3.5.3 Standard Parts. - Military standard parts shall be used whenever they are suitable for the purpose, and shall be identified by their military part number. Any part, of the same part number shall be interchangeable as applied within the system. Commercial utility parts such as screws, bolts, nuts, cottor pins, etc., may be used provided they possess suitable properties and are replaceable by the standard parts (MS and AN) without alteration and provided the corresponding standard part numbers are referenced in the parts list and, on the contractor's drawings. In the event there is not a suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

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3.6 Identification of Product. - Equipment, assemblies and parts shall be marked as specified in Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests. - Inspection and testing shall be classified as follows:

(a) Classification of Tests. Qualification tests are those tests accomplished on equipment submitted for qualification as a satisfactory product.

(b) Acceptance Test. Acceptance tests are those tests accomplished on each equipment manufactured and submitted for acceptance under a production contract.

(c) Qualification Verification Tests. Qualification verification tests are those tests conducted on product samples to verify that the design is identical to that qualified and that production units meet all the requirements of the detail specification.

4.1.1 Inspection Responsibility. - The supplier is responsible for the performance of all inspection requirements as specified herein. He may utilize his own facilities or any commercial laboratory acceptable to the government. Inspection records of the examinations and test shall be kept complete and available to the government.

4.2 Qualification. - Equipment will be considered qualified and placed on the Qualified Products List if qualification test samples (as defined below) successfully complete the qualification tests designated by Table I essentially in the order listed and upon acceptance by the qualifying activity of a maintenance plan as defined in AR-10. Equipment will be retained on the Qualified Products List as long as the qualification verification samples are delivered as required below (paragraph 4.2.3) and if these samples are of the originally qualified configuration with authorized changes incorporated and if these samples continue to demonstrate conformance to the applicable detailed specification.

4.2.1 Qualification Test Samples. - The qualification test sample shall consist of: (a) three (3) channels as described by the detailed specification which have successfully completed the acceptance tests of Table I; (b) a mating plug for each connector, electrical and hydraulic, used on each of the three (3) units; (c) two (2) sets of component wiring diagrams, outline drawings, detailed assembly drawings with parts list, and operating instructions; and (d) three (3) copies of the material to be included in the installation instructions to be packaged with each production unit and an example of how this material is to be printed, protected, and attached to the unit. Sample System No. 1 shall be

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instrumented by the manufacturer to permit the measurement of the critical component temperatures. The instrumentation used and the location of the temperature measuring points shall be subject to approval of the qualifying agency. The temperature at these points shall be monitored throughout the qualification tests of Sample No. 1.

4.2.2 Qualification Testing. - Qualification testing must be specifically authorized by the qualifying activity. At the option of the qualifying activity, any or all qualification tests required for anyone of the qualification samples (see Table 1) may be conducted on units submitted for qualification, for qualification verification, for production acceptance, or any unit delivered under a production contract for which conformance to this specification is required.

4.2.3 Qualification Verification. - Each equipment of the same part number delivered as a qualified item under this specification shall be numbered sequentially in essentially the order that is submitted for government acceptance regardless of the contract under which delivered or by whom purchased. To meet the delivery requirement for the first 50 components, 51 components must successfully complete the required acceptance tests; 50 of which are to be shipped as directed by the applicable contract(s); and one component selected at random by the Government Inspector is to be forwarded to the Naval Air Test Center (SY60), Patuxent River, Maryland 20670 (Attention: For NATC-4P7-101). To meet the delivery for each succeeding 150 components, 151 components must successfully complete the required acceptance test, 150 of which are to be shipped as directed by the applicable contract, and one component selected at random and forwarded as above. The span of serial numbers of the production units from which each verification sample was selected shall be recorded and this information included with the sample. The sample shall be shipped no later than the final unit in the production group from which selected. The units forwarded to Naval Air Test Center will be inspected and tested for conformance to the applicable detail specification and for conformance to the acceptable configuration. Tests will be conducted as shown by Table I for any one of the qualification samples at the discretion of the qualifying activity. The sample unit will be returned to the contractor with a report of the government test results at the conclusion of the design verification tests. Delivery of production units successfully passing the acceptance tests are to be accepted unless qualification is officially rescinded.

The requirements of Provisions Governing Qualification Testing, SD-6, Part II, shall pertain. Shipping authorization and instructions are to be given by the cognizant Government Inspector when the sample unit is selected. The price of testing to be borne by the contractor shall be fixed by the price list issued by the Government qualifying activity of the issue in effect on the date of the invitation for the bid which led to the production associated with the test sample. If the test

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sample is selected from group production units involving more than one contract, the price for testing will be determined by the date of the bid associated with the oldest of these contracts. A certified check (made payable to Commander, Naval Air Test Center) for the full amount of the testing is to be submitted with each unit.

4.2.4 Rejection and Retest of Qualification and Qualification Verification Samples. - Samples which have been rejected or returned to the manufacturer for any reason during qualification or qualification verification tests, may be reworked or have parts replaced to correct defects. Before resubmitting the samples, full particulars concerning the rejection and the corrective action taken by the manufacturer must be submitted in writing by the manufacturer to the qualifying activity. Tests shall not be resumed until such a report is received. Where qualification tests are conducted under the auspices of the manufacturer, the qualifying activity will be advised upon failure of a qualification sample and of the action taken by the manufacturer with regard to failure. If the manufacturer does not correct and return components rejected during qualification verification testing within three (3) days after he receives them to the qualifying activity or does not submit a program acceptable to the qualifying activity for the correction of deficiencies noted within three (3) days, the component will be disqualified.

4.2.5 Disassembly and Inspection. - At the conclusion of tests, each sample will be disassembled as necessary to inspect for wear, defects, and overheating.

4.2.6 Test Report. - A test report is required and shall be prepared in accordance with MIL-STD-831.

4.3 Acceptance. -

4.3.1 Acceptance Requirements. - Each unit produced under this specification will be accepted if:

(a) Qualified to this specification as amended by the detail specification and if the qualification is sustained in conformance with paragraph 4.2.3.

(b) Acceptance tests of Table I are successfully completed.

(c) Installation instructions approved by the qualifying activity are securely attached to each system component in such a way that they need not be removed for check out of the component prior to its installation on the aircraft.

(d) A package holding one nut, MS90415-6, for each mounting hole in the generator flange is securely attached to each generator in such a way that it need not be removed for check out of the component prior to installation on the aircraft.

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(e) All changes to equipment have been processed in accordance with MIL-STD-481.

4.3.2 Rejection and Retest of Units Submitted for Acceptance. - Units that have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

4.4 Test Conditions. - Unless otherwise specified, each test, required by Table I shall be conducted under the conditions listed in this paragraph.

4.4.1 Mounting. - The rotational axis of the generator package shall be horizontal. All other components shall be mounted in a horizontal position with base down.

4.4.2 Test Sample. - Tests are to be conducted on a complete single (non-parallel) channel. The channel must include the feeder lengths as defined by the detailed specification or lumped impedance equal to the self and mutual impedance of all feeders. The quantity of liquid coolant in the system shall be limited to that specified for service use.

4.4.3 Warm Up. - Prior to each test, the system shall be operated with the generator running at average rated rpm, delivering rated current at rated voltage, for sufficient time for the temperature of each system component to stabilize (See paragraph 6.2.11).

4.4.4 Voltage Measurement. - Whenever voltage measurements are required, line-to-neutral voltages and line-to-line voltages shall be measured at the point of regulation and recorded. Voltage measurements shall be true rms.

4.4.5 Load Conditions. - Rated channel output voltage shall be as specified in paragraph 3.3.1. Rated current shall be based on rated Channel KVA and 115 volts rms line-to-neutral at the point of regulation as defined in paragraph 6.2.1. When rated load is required for a test herein, the highest slash rating (where slash ratings are designated) shall pertain. When minimum power factor is specified for a test, the power factor of loads on each phase shall be within 0.72 to 0.78 lagging and each phase current shall be within 3% of the average magnitude of all three phase currents. Specified temperatures on all channel components are to be allowed to stabilize for each load applied during each test. The temperature shall be considered to have stabilized when the temperature rise at the points in each channel component monitored is no more than $1^{o}C$ during a period of 5 minutes.

4.4.6 Altitude and Ambient Temperature. - The altitude, ambient temperatures, and externally furnished coolant temperature shall conform to the standard conditions defined in MIL-E-81910.

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4.4.7 Generator RPM. - The generator shall be operated at average rated rpm.

4.4.8 Integral Control Power. - Full rated integral control power shall be delivered from each channel in accordance with paragraph 3.2.2 during all system tests.

4.4.9 Loading. - All system components shall be operated under rated load conditions. The system shall be turned on and off using the switch described by paragraph 3.4.1.3.

4.4.10 Thermal Isolation. - Adequate thermal insulation shall be placed between channel components and any objects with which they come in contact, and over the external surface of the generator to reduce heat transfer through these surfaces to a negligible value during thermal test. Insultation is not required at mounting face and shaft at generator input. In other tests, thermal insulation is not required on the external surfaces of the generator.

4.5 Tests. -

4.5.1 Burn-In. - Prior to conducting the tests of Table I, a 80 hour burn-in shall be performed on each channel as follows: The converter portion of the system shall be burned-in conformance with the cycle of Figure 2 of MIL-STD-781B where: -

A = 4 Hours (The system shall not be operated during 2 of 5 cycles)

B = 4 Hours

C = 4 Hours (The system shall not be operated during 2 of 5 cycles)

D = 4 Hours

High Temperature is 71°C

Low Temperature is -55°C

The equipment shall be vibrated in accordance with Paragraph 5.3.2 of MIL-STD-781B. Five cycles of the above shall be performed.

If a part in a channel component fails during the first 3 cycles of the burn-in test, it may be replaced and the test continued. If a part fails during the last 2 cycles, then it may be replaced and the test resumed at the beginning of the 4th cycle of burn-in. Upon the second failure of the same part in components undergoing qualification testing or in any sequentially serialized production run of 20 channel components, both failures shall be reported in writing to the qualifying activity within five (5) working days. The report is to identify the failure, its

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cause, and include proposed corrective action. Acceptance of production channel components may continue, at the discretion of the cognizant Government Inspector, for a period of fifteen (15) working days. Approval by the qualifying activity of the corrective action is required for acceptance of production articles after this fifteen (15) day period.

Prior to assembly, semi-conductors to be installed in generators shall successfully complete 50 cycles programmed as for the channel components above.

4.5.2 Examination of Product. -

4.5.2.1 Qualification and Qualification Verification Tests. - Each channel component and all parts thereof submitted for qualification tests or qualification verification shall be examined as the Government Inspector and qualifying activity deem necessary to determine conformance to this specification and the detail specification with respect to material and workmanship, cleanliness, standard parts, simplification, mounting flange, coupling spline lubrication, external openings, electric connections, excitation, connections, threaded parts, finish, interchangeability, modular construction marking, installation instructions, provisions for maintainability, dimensions, weight, and color.

4.5.2.2 Acceptance Test. - Each channel component sub-assembly and all parts submitted for acceptance shall be examined to determine conformance to this specification with respect to workmanship, cleanliness, lubrication, electric connections, adjustments, potentiometers and adjustable resistors, threaded parts, finish marking, installation instructions, color and consistency with qualification samples.

4.5.3 Dielectric Test. - The generator shall withstand the following test voltage at commercial frequency, applied between windings, and between each winding and frame, for the specified time. Control components shall be tested from each terminal to case. (Capacitors and electronic devices shall be disconnected if this test is likely to damage them). Leakage currents in excess of 1 milliamp shall be cause for rejection.

(a) Circuits of 50V and less: 500V for 1 minute or 600V rms for 1 second.

(b) Circuits operated above 50V: twice the rated voltage plus 1,000 rms for 1 minute, or 120 percent of the 1 minute voltage for 1 second.

(c) Capacitors and electronic devices prior to assembly shall be subjected to and shall withstand a test voltage of twice the maximum peak voltage encountered during normal operating conditions or a minimum of 100V, whichever is greater.

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4.5.4 Heating and Overload. -

(a) Demonstrate channel ability to deliver power in accordance with paragraph 3.2.

(b) With the generator package operating at minimum input rpm repeat 4.5.4.a.

(c) With generator package operating at maximum input rpm repeat 4.5.4.a.

(d) Repeat 4.5.4.a at maximum temperature.

(e) Repeat 4.5.4.a at maximum altitude.

4.5.5 Load Transients. -

(a) The ability of a single channel to meet the transient requirements of paragraphs 3.2.2, 3.3.1, and 3.3.2 including load switching, shall be demonstrated at minimum, maximum, and average generator rpm.

(b) During the transient tests, the input mechanical torque to the generator package shall be measured to demonstrate conformance to paragraph 3.3.7 at minimum, maximum, and average generator rpm.

4.5.6 Constant Load. - The regulated AC voltage and frequency, wave form distortion, phase-voltage frequency components (paragraph 3.3.1.3), and dc content of the system and the voltage for the integral control power shall be measured under the conditions listed below. For any combination of applicable temperature, load, power factor, and generator rpm, the AC voltage and frequency shall remain within limits specified by the paragraphs 3.3.1 and 3.3.2 respectively. The integral control power voltage shall remain within the limits of paragraph 3.2.2. The voltage frequency components of the system at the point of regulation shall be measured per phase for each test condition.

4.5.6.1 Voltage and Frequency Characteristics at Ambient Temperature (25 + 150°C). -

(a) 1. With minimum rated input shaft rpm to the generator package, the AC load shall be varied in steps from no load to 150%. The load shall be connected using the manual control (paragraph 3.4.1.3). Sufficient data shall be taken to determine the steady state voltage and frequency regulation per phase over a load range of from zero to 150%.

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2. With minimum rated input shaft rpm to the generator package, the AC load shall be varied from a per phase load of zero to 100% load in steps wherein the difference of load per phase is 1/3 of the phase rating. Sufficient data shall be taken to determine the steady state AC voltage and frequency regulation per phase over a load range of from zero to 100%.

(b) At average rated generator rpm repeat the tests of paragraph 4.5.6.1(a).

(c) With maximum rated input rpm to the generator package, repeat the tests of paragraph 4.5.6.1.a.

(d) At zero AC output current, the generator package input rpm shall be gradually varied from minimum to maximum and returned to minimum rated rpm. Data shall be recorded at 10 equally spaced intervals of rpm and at any point where the limits of paragraphs 3.2.2, 3.3.1, and 3.3.2 are exceeded.

(e) At 50 percent rated AC current, the preceeding test shall be repeated at minimum, average, and maximum power factor.

(f) At 100 percent rated AC current, the preceeding tests shall be repeated.

4.5.6.2 Voltage and Frequency Characteristics at -55°C. - Repeat the test of paragraph 4.5.6.1(a).1 at -55°C.

4.5.6.3 Voltage and Frequency Characteristics at the Maximum Ambient Temperature(s). - Repeat the test paragraph 4.5.6.1(a).1 at the maximum temperature(s) indicated by paragraph 3.4.1.2.

4.5.6.4 Split Synchronized Operation. -

4.5.6.4.1 Synchronization of VSCF Channels. - Two VSCF channels shall be operated split synchronized as follows to demonstrate that the requirements of paragraph 3.3.5 are met.

(a) With the two channels at no load and then when simultaneously delivering 0, 20, 60, 100, and 150 percent load.

(b) With channel 1 supplying full load and channel 2 delivering 0, 20, 60, 100 and 150 percent full load.

(c) With channel 2 supplying full load and channel 1

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delivering 0, 20, 60, 100 and 150 percent full load.

4.5.7 Standard Test for Electric Power Equipment. - Conduct tests in accordance with Table I of specification MIL-E-81910. The channel is to be checked in accordance with paragraph 4.5.6.1(b) before and after the required tests and in accordance with paragraph 4.5.6.1(b) where a performance test is required during the test of specification MIL-E-81910. The class of equipment, defined under paragraph 4.8 of MIL-E-81910 is to be identified by the detail specification. In addition, conformance to paragraph 3.4.2.3 is to be determined during environmental tests.

4.5.8 Efficiency. - The power output and input of the channel shall be measured at full load at a power factor of 0.7 lagging and 0.95 lagging with an input shaft speed to the generator of minimum, average, and maximum rated rpm respectively. The efficiency shall be calculated under these conditions and shall not be less than that specified by paragraph 3.3.3. Sufficient additional data shall be taken to determine losses over the load range at each of these three generator package speeds and to demonstrate full compliance to paragraph 3.3.3.

4.5.9 Heat Rejection. - Heat rejection rate from each of the system components shall be tested for conformance to paragraph 3.3.3 in conjunction with the efficiency test paragraph 4.5.

4.5.10 Factors Affecting Voltage and Frequency Regulation. -

4.5.10.1 Affect of Individual Environments on Regulation. -

4.5.10.1.1 Ambient Temperature. - The ambient temperature shall be decreased from $25^{0} + 15^{0}$ to -55^{0}C; increased to the maximum ambient temperature(s) indicated by paragraph 3.4.1.2 and then decreased to $25^{0} + 15^{0}$ C. Each temperature shall be held for at least one hour or until the voltages and frequency stabilize as indicated by readings taken at 10 minute intervals. The ability of the channel to deliver rated AC and DC outputs continuously at ambients between -55^{0}C and the maximum ambient temperature(s) indicated by paragraph 3.4.1.2 shall be demonstrated and the values of the voltages and frequencies recorded. Tests may be made at any ambient temperature between the limits specified at the option of the Government Inspector or testing activity.

4.5.10.1.2 Power Factor. - The load power factor shall be decreased from 0.95 lagging to 0.75 lagging. Tests will be made at any other power factors required by the detailed specification. Each load power factor shall be held until the voltage and frequency stabilize. The ambient temperature at each altitude shall be within 5^{0}C of the maximum specified by paragraph 3.4.1.2.

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4.5.10.1.3 Altitude. - The pressure altitude shall be varied from approximately sea level to conditions of maximum altitude and temperature indicated by paragraph 3.4.1.2. Tests shall be made at intermediate altitudes of one third and two thirds the maximum altitude designated by the detailed specification. There shall be no evidence of corona or harmful arching under the altitude conditions. Each altitude shall be held until the voltage and frequency stabilize. The ambient temperature at each altitude shall be within 5°C of the maximum specified by paragraph 3.4.1.2.

4.5.10.1.4 Position. - During the course of tests under this section, static system components shall be rotated by increments of no more than 90 degrees about each major axis, except the vertical axis and the requirements of paragraphs 3.4.2.7 and 3.4.3.3 demonstrated.

4.5.10.1.5 Warmup. - Only the maximum voltage and frequency deviation values obtained from the tests below shall be used in computing total deviation:

(a) The channel shall be soaked at an ambient temperature of -55°C for a period of at least 8 hours during which time no power shall be connected to any component of the channel. At the end of this period, without changing the ambient temperature and with the channel switch in the "ON" position, the generator shall be brought up to minimum rated rpm and rated AC load and rated integral power load conditions shall be applied. Readings of the voltages and frequency at the point of regulation shall be recorded until the voltage and frequency stabilize. The voltage and frequency limits shall not exceed the limits of paragraph 3.3.1.1 and 3.3.2.1. Readings taken during the first 25 seconds shall not be used in the computation of the total deviation.

(b) Repeat the tests of paragraph 4.5.10.1.5.a with no load on the unit.

(c) Repea the tests of paragraph 4.5.10.1.5.a at the maximum ambient temperature(s) indicated by paragraph 3.4.1.2.

(d) Repeat the tests of paragraph 4.5.10.1.5.b at the maximum ambient temperature(s) indicated by paragraph 3.4.1.2.

For both voltage and frequency, the 5 values of maximum positive deviations recorded from tests 4.5.10.1.1 through 4.5.10.1.5 above shall be added together and the 5 values of negative deviations shall be added together. To these sums add respectively the maximum positive and the maximum negative deviation owing to changes in load and input generator speed at room atmosphere which were recorded during the test specified in 4.5.6.1.a, b, and c. The total positive and the total negative deviations thus obtained shall be

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added algebraically to the standard values of voltage and frequency separately and the system shall be considered acceptable if the voltage and frequency values thus obtained remain within the limits specified by paragraphs 3.3.1.1 and 3.3.2.1.

4.5.10.2 Regulation at Most Adverse Indicated Conditions. - If the channel is found to be not acceptable under the total deviation test specified in paragraph 4.5.10.1, further tests shall be run to determine regulation at the combination of the most adverse indicated conditions as follows:

(a) The most adverse indicated conditions of temperature, altitude, operating position, input shaft speed, load power factor, and magnitude of load to be used for this test shall be those conditions in the tests specified in paragraph 4.5.10.1 and 4.5.6.1.a, b, or c where the channel was found to have the maximum positive voltage deviations. The channel components shall be placed in the most adverse indicated operating position and soaked for at least 8 hours (not operating and with no circuits energized) at the most adverse indicated ambient temperatures after which the pressure altitude shall be adjusted to the most adverse indicated value for the system. During this test, the temperature-altitude conditions shall not exceed the limits specified by paragraph 3.4.1.2. The generator shall then be brought up to minimum rpm with the control switch in the "ON" position. The most adverse indicated load shall be immediately applied. Readings of the voltages and frequency shall be recorded immediately after starting and until they stabilize. All readings after the first 25 seconds shall be within the specified limits.

(b) The test, paragraph 4.5.10.2.a, shall be repeated for conditions found to produce maximum negative voltage deviations.

(c) Repeat the tests of paragraph 4.5.10.2.a for conditions found to produce maximum positive frequency deviations.

(d) Repeat the tests of paragraph 4.5.10.2.a to produce maximum negative frequency deviations.

If the voltage and frequency remain within the specified limits during the above tests, the channel shall be considered to have satisfactorily fulfilled the requirements of the test for factors affecting regulation even though the regulation values computed by using total deviations specified in paragraph 4.5.10.1 did not remain within the specified limits.

4.5.11 Operation of Manual Control. - During the tests provided under

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paragraph 4.5.14 the channel shall demonstrate compliance to paragraph 3.4.1.3. If paralleling or synchronizing is required by the detail specification, the systems shall first be run through this test individually and then in parallel and/or synchronized as required.

4.5.12 Operation of Protection Devices. -

4.5.12.1 Qualification and Qualification Verification. - An analysis shall be made of the channel to determine which malfunctions will result in tripping of the protective devices or could result in the system not meeting the requirements of paragraph 3.3. Sufficient tests shall be conducted to determine the validity of the analysis and to demonstrate the functioning of the protective devices specified by paragraph 3.3.1, 3.3.2, 3.4.2.5, 3.4.3.1 and 3.3.4.3. Any condition within the specified operating ranges of the channel which could permit a fault to remain on the bus undetected shall be tested and the probability of its occurrence under service conditions analyzed. After it is determined which faults will result in tripping the channel off the bus being powered by the system, samples of these faults shall be applied to the channel to determine that the protective devices operate within the specified limits. The protective functions shall be tested at 25°C, -55°C, and the maximum temperature specified in paragraph 3.4.1.2 to determine that no false tripping will occur and that channel protection requirements are met at all specified temperatures.

4.5.12.2 Acceptance. - Sufficient data shall be taken to demonstrate the compliance of protective devices specified by paragraph 3.3.1, 3.3.2, and 3.4.3.1 at an ambient temperature of 25 ± 15°C.

4.5.13 Overspeed. - This test shall be made on the generator package while hot as a result of testing, and shall be made at no load at a pad speed as defined by paragraph 3.4.2.2. It shall demonstrate its ability to operate under overspeed conditions for 5 minutes without impairing subsequent specified performance. Immediately following this test, the channel shall pass the test specified in paragraph 4.5.6.1.b.

4.5.14 Performance and Endurance. - The channel shall operate failure free for a minimum time of 2000 hours during this test (as an option two channels may be used for this test with each completing one-half of the test requirements). Conformance to paragraph 3.4.1.13 shall be demonstrated to a 90% confidence level by completion of this test without failure. The channel(s) shall meet the performance requirements contained herein during all phases of this test. Cooling to the various channel components shall be as defined by the detail specification and shall be exercised on a cyclic basis throughout the test.

4.5.14.1 Single Channel Test. Time shall be accumulated by completing 105 of the cycles shown below for room ambient operation plus 20 of the

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temperature cycles shown in Figure 2 of MIL-STD-781B. The channel shall be shut down, generator not turning, for a minimum of 4 hours after each 5 cycles of operation. The shut down time shall not be considered as test time. Five temperature cycles, Figure 2 of MIL-STD-781B with A = B = C = D = 4 hours, a high temperature of 80°C and a low temperature of -55°C, shall be accomplished on the channel after each completed 25 cycles of room ambient operation. The channel shall be operated at no load throughout the temperature cycle with the temperature cycling done on only the converter.

Room Ambient Operation

Generator Input Speed	% Load On System	Integral Control Power Load (%)	Operating Time (hrs)
Min	0	0	1
Min	25	100	1
Min	100	100	1
Max	100	100	2
Ave	100	100	2
Ave	50	100	4
Max	50	100	2
Ave	100	100	2
Min	50	100	1

The following overload and fault conditions shall be imposed and shall be considered as test time. These tests shall be distributed over the whole testing period.

a. 200% AC load at 0.75 power factor for a duration of five seconds, 50 times. 200% DC load, 50 times.

b. 150% AC load at 0.75 power factor for a duration of five minutes, 50 times. 150% DC load, 50 times.

c. AC single phase and two phase line-to-line and line-to-neutral short circuits, as well as DC short circuits on the integral control power circuit shall be repeated 3 times for a minimum of five seconds for each of the generator input speed conditions.

4.5.14.2 System Test. - When paralleling, split synchronized and/or no break operation is required by the detail specification the applicable tests defined by paragraph(s) 4.5.14.2.1, 4.4.14.2.2, and/or 4.5.14.2.3 shall be conducted.

4.5.14.2.1 Parallel Operation. - The channel shall be paralleled with a second channel during four of the required 105 cycles

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of room ambient operation defined by paragraph 4.5.14.1. The generators shall be operated from separate unsynchronized driving pads. During the intervals when the load is 50% or less, each individual channel shall be switched off once for 5 minutes and then on while the other channel is operating. No nuisance trips or interruptions of power to the load shall occur. System conformance to paragraphs 3.3.1 and 3.3.2, shall be demonstrated. The following overload and fault conditions shall be imposed on the common bus and shall be considered as test time.

a. A load equal to the rated load current of one channel multiplied by twice the number of systems operating in parallel shall be supplied by the systems for a duration of five seconds, two times.

b. A load current equal to the rated load current of one system multiplied by 1.5 times the number of systems operating in parallel shall be supplied by the system for a duration of five minutes, two times.

c. Single phase and two phase line-to-line and line-to-neutral short circuits shall be applied and held for five seconds, once at each different generator input speed.

4.5.14.2.2 Split-Synchronized Operation. - The channel shall be operated split-synchronized with a second channel for four of the required 105 cycles of room ambient operation defined by paragraph 4.5.14.1. The generators shall be operated from separate unsynchronized driving pads. System conformance to paragraphs 3.3.1 and 3.3.2 shall be demonstrated. Each individual channel shall be switched off for 5 minutes and then back on while the other channel is operating, 5 times for each channel. No nuisance tripping shall occur. The following overload and fault conditions shall be imposed on each of the channels alternately and considered as test time. These tests shall be distributed over the whole testing period. The channels shall remain synchronized.

a. A load current equal to twice the rated channel current shall be supplied by each channel for five seconds, twice for each channel.

b. A load current equal to 1.5 times the rated channel current shall be supplied by each channel for a duration of five minutes, twice for each channel.

c. Single phase and two phase line-to-line and line-to-neutral short circuits shall be applied to each channel for a minimum of five seconds, twice for each channel.

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4.5.14.2.3 No-Break Transfer Operation. - Conformance to paragraph 3.3.6 shall be demonstrated by transferring loads from each channel to and from systems of a different basic design (simulating ground power), 25 times. These tests shall be distributed over the whole testing period and shall be considered as test time.

4.5.15 Vibration Acceptance Tests. - No vibration test is required for generators at time of acceptance. All system control components shall be subjected to a vibration excursion from 5 to 2000 Hz and return over a 15 minute period. The vibration shall be at 0.036 inch double amplitude or an applied acceleration of 10G whichever is the lower value. The direction of vibration shall be along the "X" axis for the first unit, along the "Y" asix for the second unit, and "Z" axis for the third, "X" axis for the fourth and etc. The axis designations are as defined in MIL-E-81910.

4.5.16 Effects of Fluids. - During the course of other tests or based on materials test results, conformance to paragraph 3.4.2.6 shall be determined. Conformance to paragraph 3.4.1.18 shall be demonstrated.

4.5.17 Flight Service Evaluation. - A flight evaluation in excess of 100 hours shall be conducted on two systems during qualification. If paralleling is required, the systems may be operated in parallel at the option of the qualifying activity. Qualification verification samples may be flight tested at the option of the qualifying activity.

4.5.18 Fluid Contamination Test. - Prior to delivery and with the final filter installed, channel components shall be operated for one hour and the oil then checked for contamination in accordance with Naval Air Systems Command, Aviation Hydraulics Manual 01-1A-17, Section III, using the contamination analysis Kit Part Number 57C414 (Fig 3-1). The maximum acceptable particule level for this equipment is Class 5 as determined by NAVAIR Manual 01-1A-17, Section 16. If the contamination level is unacceptable the unit may be operated for a second one hour period with a clean filter. If the above criteria is met after the second hour the unit is to be disassembled and the source of contamination identified before resubmitting for test.

4.5.19 Maintainability Demonstration. - Compliance to paragraph 3.4.1.17.3 shall be demonstrated by performing three identical successive replacement actions by the same maintenance technician on each channel component. The duration of each replacement action shall be measured and shall include all active repair time from initiation of disassembly to completion of reassembly, but excluding time expended in verification of functioning. The sum of the durations of the second and third replacement actions shall not exceed the time specified in paragraph 3.4.1.17.3.

4.5.20 Disassembly and Inspection. - This test shall consist of disassembling the system components and inspecting for defects and wear.

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Such inspection (of at least two of the three qualification samples or the qualification verification samples) shall be done during overhaul in accordance with the overhaul instructions, by personnel with the designated skill levels, to demonstrate compliance with paragraph 3.4.1.17.2. The contractor shall supply all necessary special support equipment for the demonstration. Equipment shall then successfully pass the Acceptance Tests of Table I. Compliance to paragraph 3.4.1.17 shall be determined during this inspection.

5. PREPARATION FOR DELIVERY. -, The packaging, packing, and marking requirements specified by MIL-STD-794 shall apply.

6. NOTES

6.1 Intended Use. - The channels covered by this specification are intended for use on aircraft as a continuous source of ac electrical power and dc control power.

6.2 Definitions. - The following paragraphs define or give reference defining terms used in this specification to assist in clarity.

6.2.1 Rated Current. - The per phase current at full rating is defined as $\frac{\text{Rated KVA X 1000}}{345}$.

6.2.2 Rated Load. - Rated load is the static linear impedance required to draw rated current under a steady state voltage of 115 volts on each of the 3 output phases.

6.2.3 Distortion. - Distortion is the rms value of the ac waveform exclusive of the fundamental and includes the components resulting from amplitude modulation as well as harmonic and non-harmonic components.

6.2.4 Distortion Factor. - The distortion factor is the ratio of the distortion to the rms value of the fundamental component.

6.2.5 Distortion Spectrum. - The distortion spectrum qualifies distortion in terms of the amplitude of each frequency. The distortion spectrum includes the components resulting from amplitude and frequency modulation as well as harmonic and non-harmonic components of the ac waveform.

6.2.6 Ripple Amplitude. - The ripple amplitude is the maximum value of the difference between the average and the instantaneous values of a pulsating unidirectional wave.

6.2.7 Feeders. - Feeders are any power carrying conductors (or groups of conductors) which transmit power between the generator and the main line contactor. Generator feeders are those which transmit power from the generator to the converter. The 400 Hz feeders are those which transmit power

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from the converter to the main line contactor.

6.2.8 Locked Rotor Impedance. - The per phase locked rotor impedance of a motor is the line-to-neutral voltage divided by the line current measured while the rotor is locked, i.e. not allowed to rotate.

6.2.9 Integral Control Power. - This is a power supply for channel control functions which forms an essential and integral part of the system and independent of any source of energy other than the system itself.

6.2.10 Module. - A module is an integral sub-assembly which is easily removable from and replaceable in the assembly as a unit.

6.2.11 Stabilized Temperature of Components. - The temperature of a component shall be considered to have stabilized when the temperature rise at the points monitored is no more than $1^{o}C$ during a period of 5 minutes.

6.2.12 Channel Component. - Any element of the channel which is removable from the aircraft as a unit.

6.2.13 Generator. - The generator includes all channel elements attached to the generator when installed on the aircraft.

6.3 Detailed Specification Inclusions. - The following minimum information is to be included in the detailed specification.

Frequency Regulation, Class (3.3.2.1)
Feeder Description (3.2.1)
Environmental Requirements (3.4.1.2)

Channel Requirements

KVA Rating (3.2.1)
Integral Control Power Capacity (3.2.2)
Whether single, parallel, split-synchronized operation, and/or no-break power transfer required. (3.3.4, 3.3.5 and 3.3.6)

Generator Requirements

Maximum Efficiency and Maximum Heat Rejection (3.3.3)
Input RPM Range for the Generator Package (3.4.2.2)
Maximum Input Torque Allowable (3.3.7)
Operating Position (3.4.2.7)
Shear Section Load Limits (3.4.3.5)
Cooling conditions (3.4.1.2)
Coolant Quantity (4.4.2)
Direction of Input Rotation
Mounting and QAD Details
Weight and Overhung Moment (3.4.1.13)

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Envelope
External Electric and Plumbing Connection
Nameplate Data
Disconnect Activation Means (if not automatic)
Manhours Required for Overhaul and Personnel Competence Level (3.4.1.17.2)
Replacement Action, Manhours Required, and Personnel Competence Level (3.4.1.17.3)

Requirements for System Components other than the Generator

Minimum Efficiency and Maximum Heat Rejection
Frequency Regulation Class (3.3.2.1)
Sensing Elements (3.4.3)
Component Weights
Component Envelopes and Mounting Details
External Electric and Plumbing Connections
Max. Cont. Heat Rejection for Each System Component
Manhours Required for Overhaul and Personnel Competence Level (3.4.1.17.2)
Replacement Action, Manhours Required, and Personnel Competence Level (3.4.1.17.3)
Coolant Quantity (4.4.2)

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TABLE I

TEST	PARA. NO.	ACCEPT. TEST	Qualification Tests TEST CHANNEL NO. 1	TEST CHANNEL NO. 2	TEST CHANNEL NO. 3
Burn-In	4.5.1	X			
Examination of Product	4.5.2.1		X	X	X
	4.5.2.2	X			
Dielectric Test	4.5.3	X			
Heating and Overload	4.5.4		X		
Load Transients	4.5.5		X	X	X
	4.5.5(a)	X			
Constant Load	4.5.6		X		
	4.5.6.1			X	X
	4.5.6.1(a)1/ (b)1/(c)1	X			
Standard Tests	4.5.7		X[1]	X[1]	X[1]
Efficiency	4.5.8		X	X	X
Heat Rejection	4.5.9		X		
Factors Affecting Voltage & Frequency	4.5.10		X		
Operation of Manual Control	4.5.11			X	

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TABLE I

TEST	PARA. NO.	ACCEPT. TEST	Qualification Tests TEST CHANNEL NO. 1	TEST CHANNEL NO. 2	TEST channel NO. 3
Operation of Protective Devices	4.5.12.1		X	X	
Operation of Protective Devices	4.5.12.2	X			
Overspeed	4.5.13	X	X		
Performance & Endurance	4.5.14		X[2]	X[2]	X[2]
Vibration, Acceptance	4.5.15	X			
Effects of Fluids	4.5.16		X[3]	X[3]	X[3]
Flight Service Evaluation	4.5.17		X[3]	X[3]	X[3]
Fluid Contamination Test	4.5.18		X[3]	X[3]	X[3]
Maintainability Demonstration	4.5.19				X
Disassembly & Inspection	4.5.20		X[3]	X[3]	X[3]

Notes: 1. Sample selection is to be as designated by MIL-E-81910, Table I.

2. Single channel tests shall be conducted on test channel No. 2. Dual channel tests shall be conducted on test channel No. 2 plus either of the other two channels.

3. Any two of the tests channels may be selected for this test.

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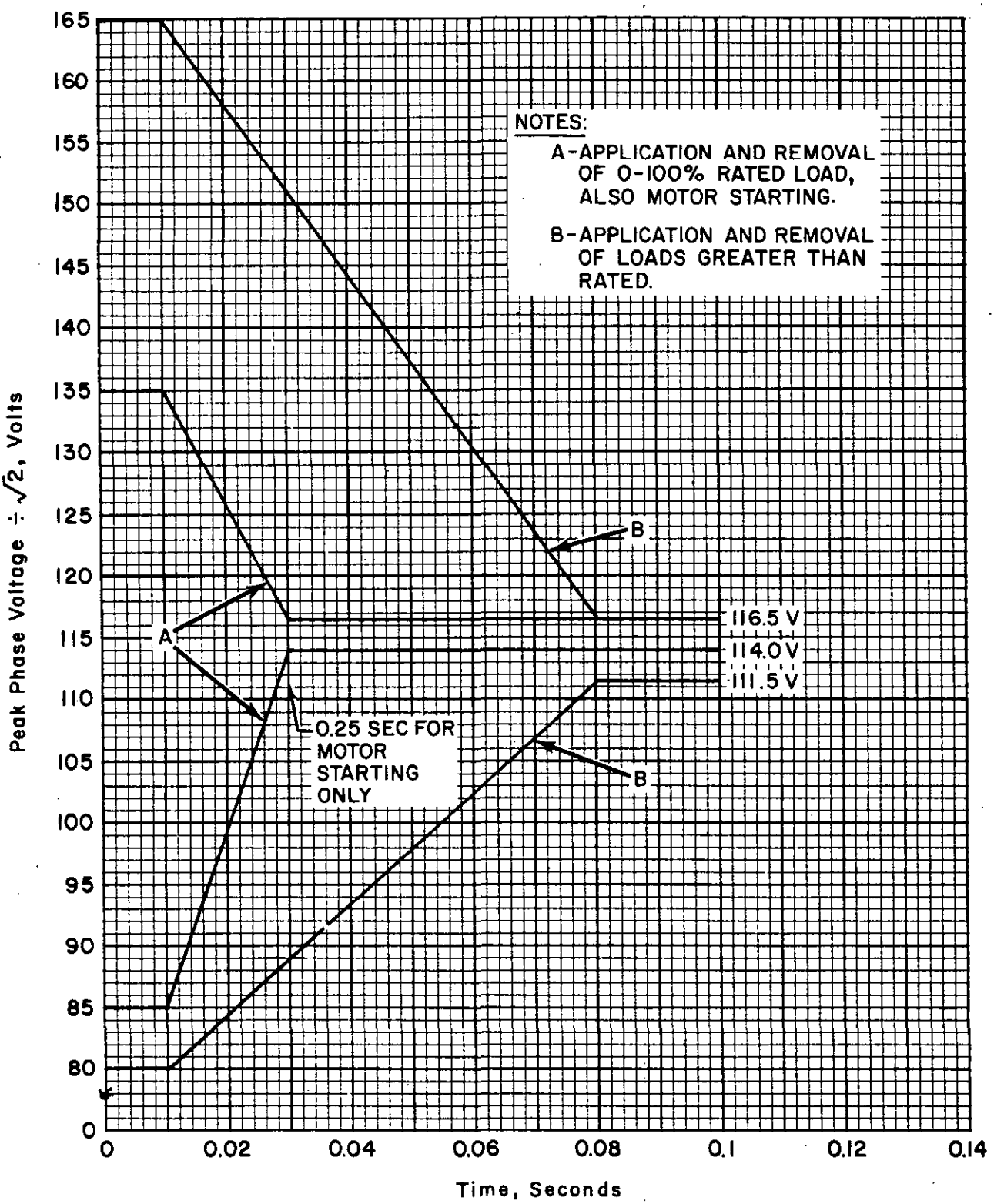


Figure 1 - Voltage Transient Limits

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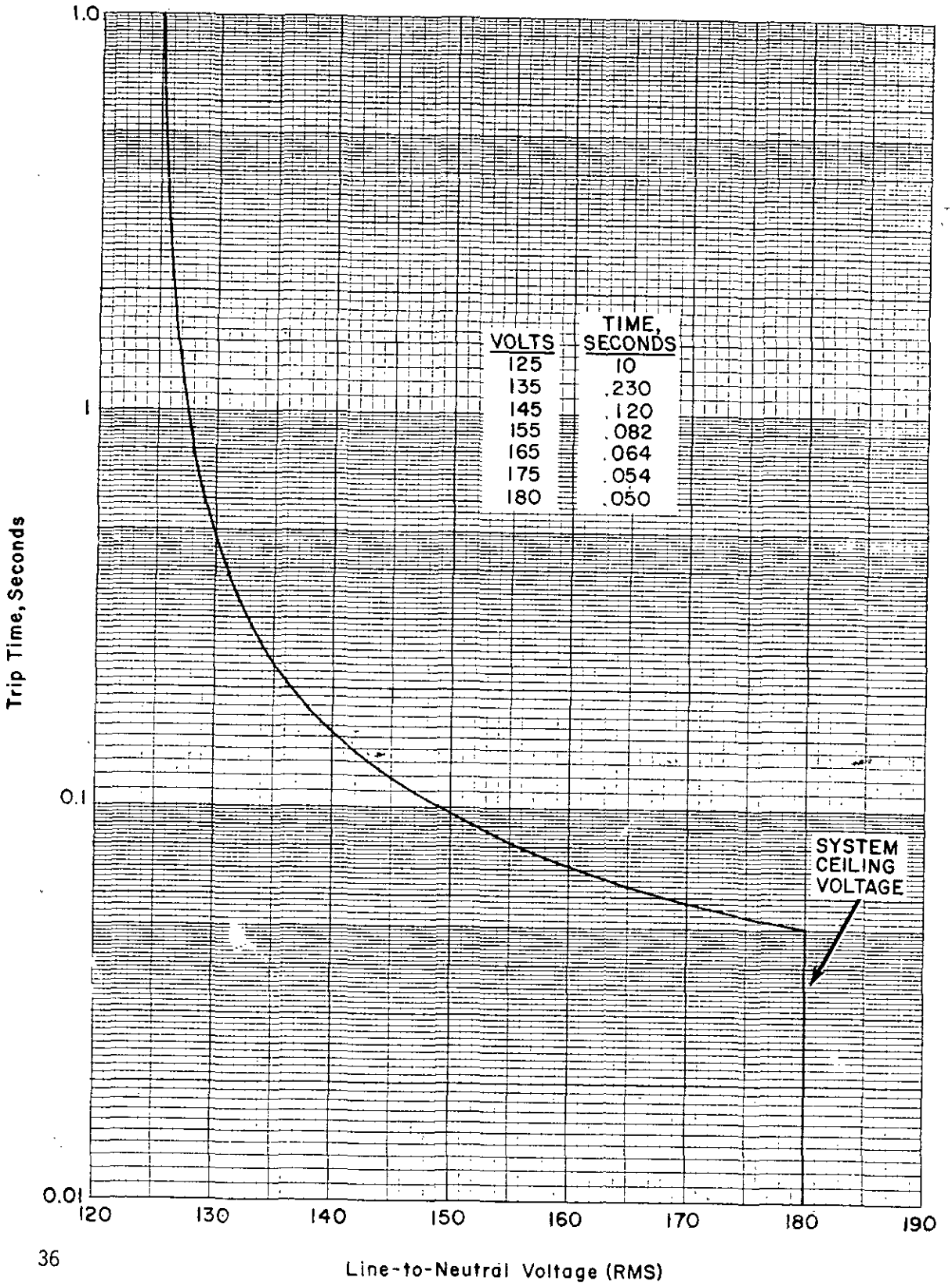


Figure 2 - Overvoltage Protection Trip Time

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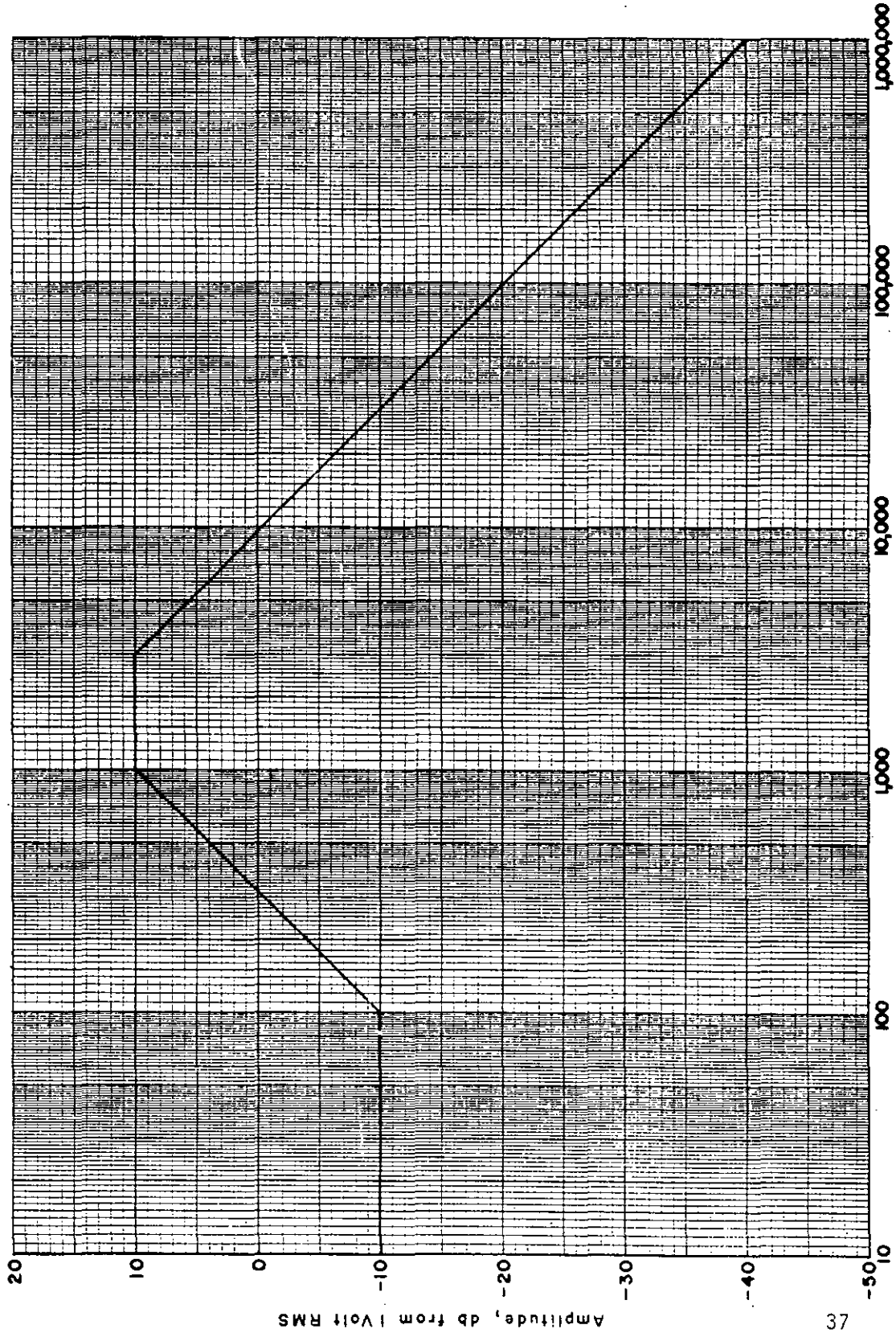


Figure 3 - Envelope of AC Distortion Spectrum

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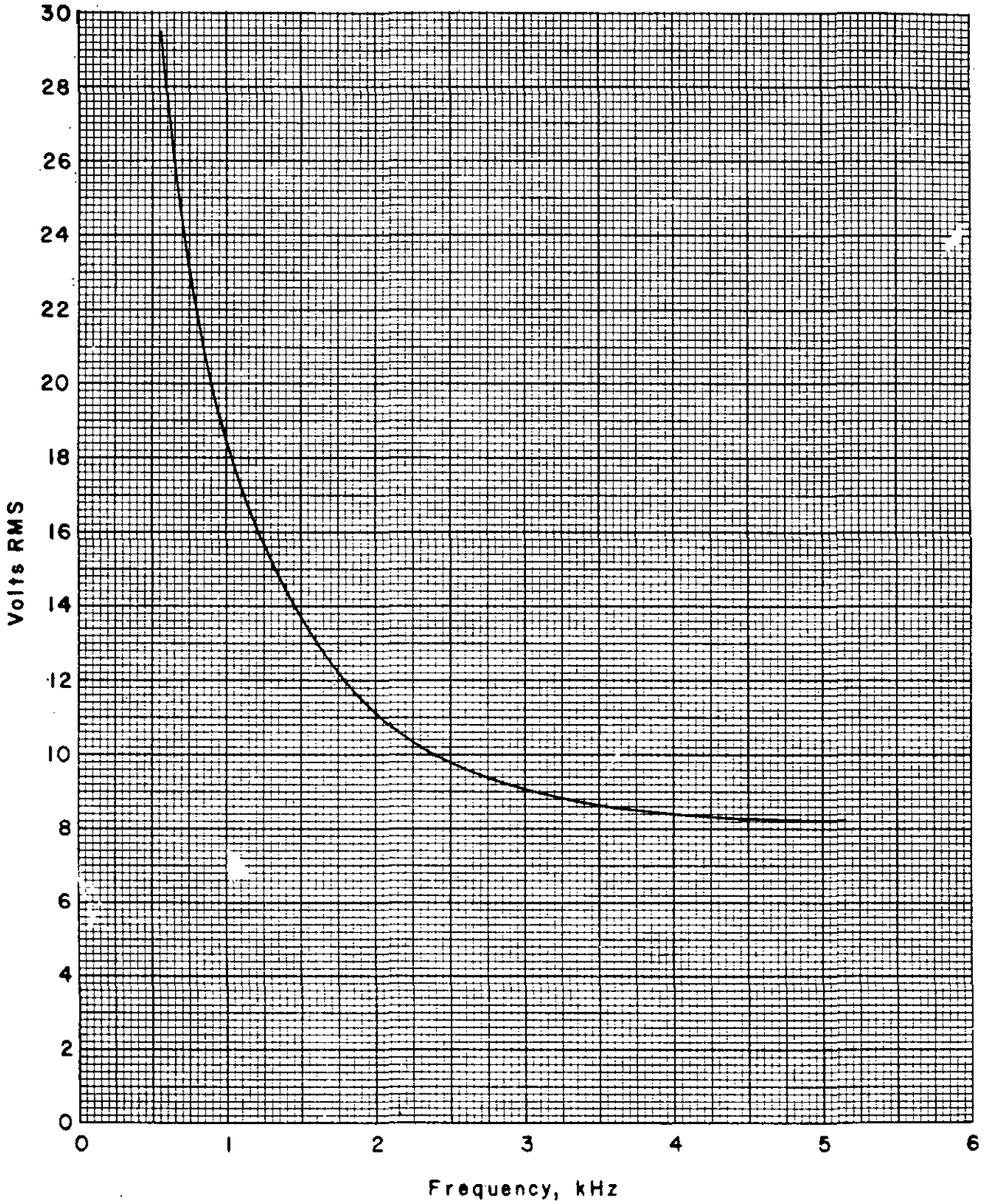


Figure 4 - AC Distortion Protection

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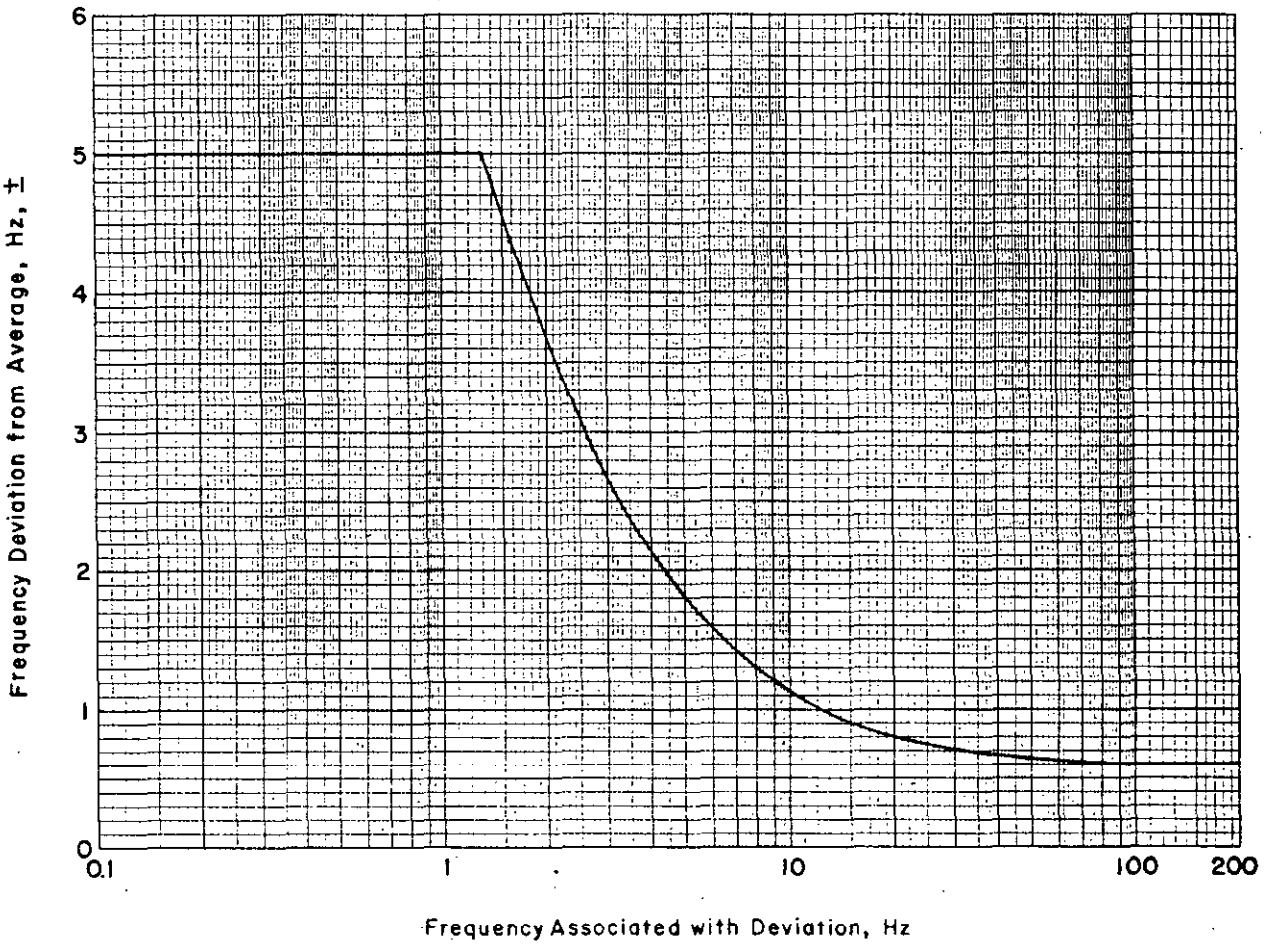


Figure 5 - Limits of Frequency Deviation

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DOCUMENT IDENTIFIER AND TITLE: MIL-E-23001B(AS) Electric Generating System, Variable Speed Constant Frequency Aircraft, General Specification for

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

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